

FSK: Tutorial Problem Numbers

2025

1 Chapter 1

Chapter 1 Physics and Measurement

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

P1.4 $m = \frac{4\pi\rho(r_2^3 - r_1^3)}{3}$

P1.9 (a) Not dimensionally correct (b) Dimensionally correct

P1.12 Number of pages are insufficient. If you do not have a physical copy of the textbook, a page is about 21 cm by 28 cm

P1.13 $r_{\text{Al}} = 2.86 \text{ cm}$

P1.14 $r_{\text{Fe}}(1.43)$

P1.31 10^{11} stars

P1.35 (a) $478 \text{ cm}^3/\text{s}$ (b) 0.225 cm/s (c) explanation